UBNetDef F25 Risk Analysis and Management

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Table of contents

Table Of Contents

Memo: 3
Executive Summary 4
Technical Findings: 5
Section1: PII Assessment 8
SectionI (continued) 9
Section 2: Wazuh SIEM Review 10
Work Cited 12
Appendix 13

Memo:

To: David Murray, Chief Executive Officer, UBNetDef
From: Ethan Messer, Security Engineer, UBNetDef SysSec
Date: November 16, 2025
Subject: Risk Analysis and SIEM Countermeasure Adjudication Report

CEO Murray,

The requested informational report assessing the CIO's plan to deploy a SolarWinds Security Information and Event Manager (SIEM) on the UBNetDef Wiki infrastructure website is attached. This report will look at the actual user data that the Wiki has collected, evaluates whether sensitive PII is present or not, and makes a risk based decision about whether the proposed expense is warranted. The report also describes a low-cost SIEM substitute that preserves organizational revenue while retaining security visibility.

Please let me know if further analysis or clarification is needed.

Respectfully,
Ethan Messer
Security Engineer, UBNetDef SysSec

Executive Summary

According to UBNetDef SysSec, there is no need for the suggested SolarWinds SIEM implementation on the UBNetDef Wiki network.

According to our analysis of the UBNetDef Wiki, users are only asked for their usernames, hashed passwords, and optional fields for their email address or real name. The system does not contain any sensitive PII, including government IDs, financial information, or Social Security numbers. The CIO's worries about significant legal or regulatory exposure are reasonable given the website's limited storage of low-risk data, but they are exaggerated in light of the data that is actually gathered. Wazuh is a suitable and affordable alternative SIEM solution for the UBNetDef Wiki, according to UBNetDef SysSec.

Without the expensive licensing and monthly costs, Wazuh now offers the same essential security features as SolarWinds, including log monitoring, alerts, file-integrity checks, and intrusion detection. According to UBNetDef SysSec, two Security Engineers can install and configure on Wazuh in three to four hours, and it can operate on the current Wiki infrastructure without the need for additional hardware. This strategy maintains the Wiki's profitability after launch while offering robust security monitoring.

Technical Findings:

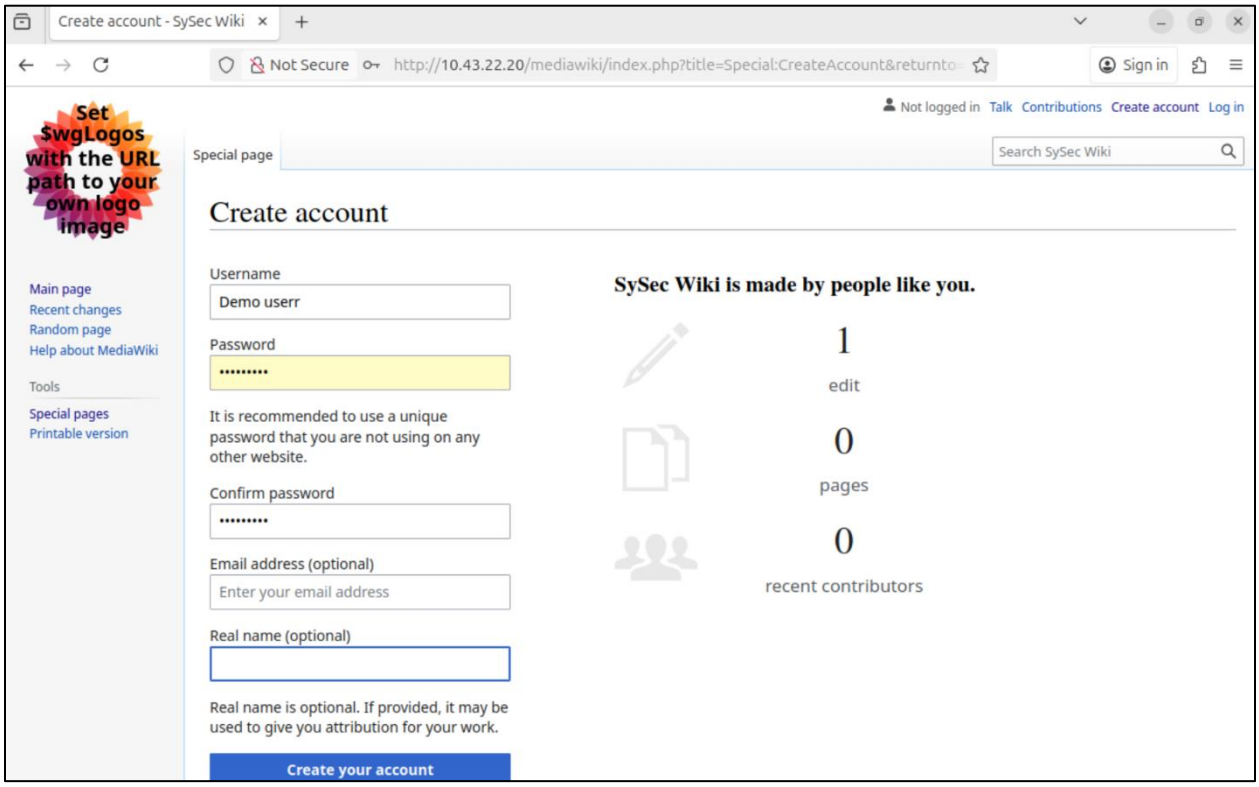

Figure 1 — MediaWiki Account Creation Page, This figure shows the user registration form, which only collects a username, password, and optional fields for email address and real name. No fields for Sensitive PII (SPII) exist in the registration workflow.

MediaWiki account creation page display the exact fields collected during user registration; username, password, optional email address, and optional real name. This demonstrates that the UBnetDef Wiki does not request or store any sensitive PII during the registration process.

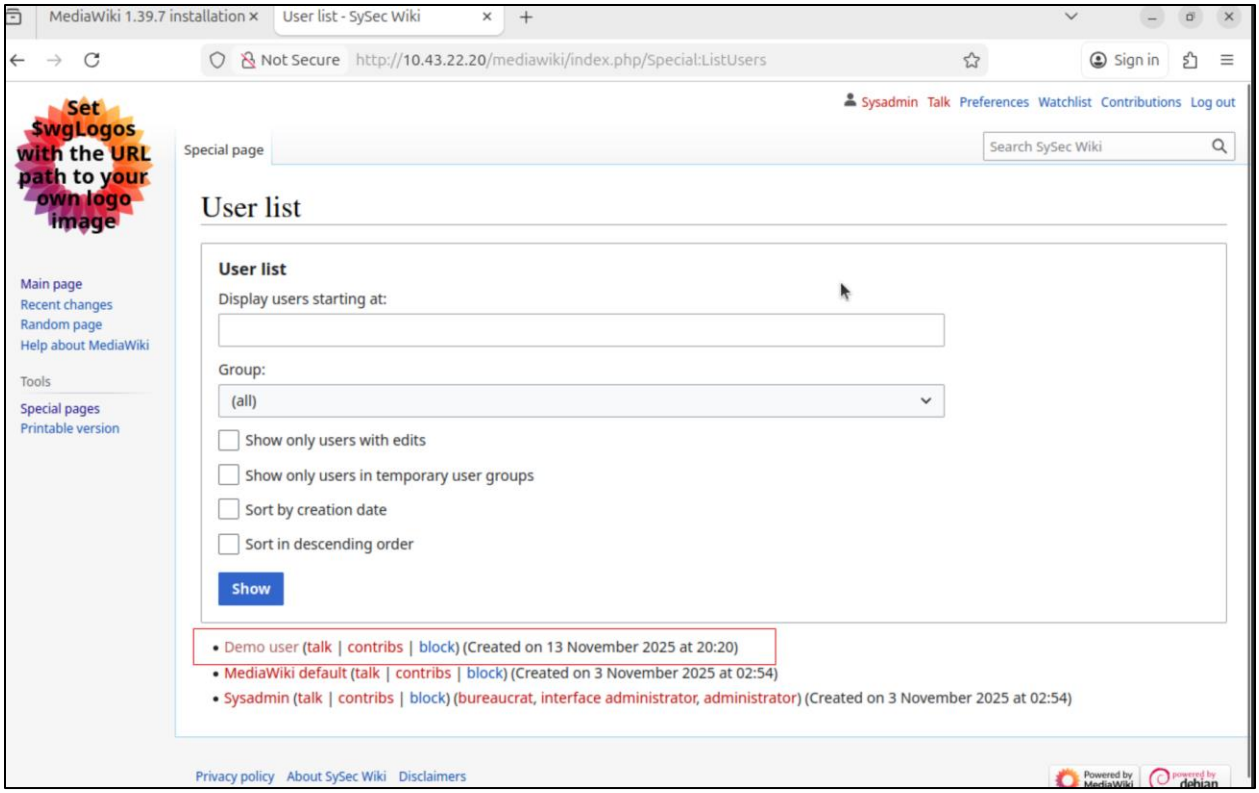

Figure 2- Mediawiki Administrative User List, 2This figure displays the administrator's view of registered accounts. Only usernames and basic metadata (such as creation timestamps) are visible; no SPII can be accessed through the administrative frontend

MediaWiki administrator view of registered user accounts, showing that only usernames and basic account metadata (such as creation timestamps) are accessible. No sensitive PII is displayed through the administrative frontend.

```
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [mywiki]> DESCRIBE user;
+--------------------------+-------------------+------+-----+-------------------------------+----------------+
| Field                    | Type              | Null | Key | Default                       | Extra          |
+--------------------------+-------------------+------+-----+-------------------------------+----------------+
| user_id                  | int(10) unsigned  | NO   | PRI | NULL                          | auto_increment |
| user_name                | varbinary(255)    | NO   | UNI |                               |                |
| user_real_name           | varbinary(255)    | NO   |     |                               |                |
| user_password            | tinyblob          | NO   |     | NULL                          |                |
| user_newpassword         | tinyblob          | NO   |     | NULL                          |                |
| user_newpass_time        | binary(14)        | YES  |     | NULL                          |                |
| user_email               | tinyblob          | NO   | MUL | NULL                          |                |
| user_touched             | binary(14)        | NO   |     | NULL                          |                |
| user_token               | binary(32)        | NO   |     |                               |                |
| user_email_authenticated | binary(14)        | YES  |     | NULL                          |                |
| user_email_token         | binary(32)        | YES  | MUL | NULL                          |                |
| user_email_token_expires | binary(14)        | YES  |     | NULL                          |                |
| user_registration        | binary(14)        | YES  |     | NULL                          |                |
| user_editcount           | int(10) unsigned  | YES  |     | NULL                          |                |
| user_password_expires    | varbinary(14)     | YES  |     | NULL                          |                |
+--------------------------+-------------------+------+-----+-------------------------------+----------------+
15 rows in set (0.002 sec)

MariaDB [mywiki]> SELECT * FROM user;
+---------+-----------------+-----------------+-----------------------------------
```

Section1: PII Assessment

Assess whether or not SPII can exist on UBNetDef Wiki

SPII is not allowed on the UBNetDef Wiki, according to the results of the administrative interface, backend database, and MediaWiki registration procedure. Email address and real name are optional fields on the registration form, which only asks for a username and password. There are no fields for any type of SPII, including dates of birth, financial account information, driver's license numbers, Social Security numbers, biometric information, medical information, or other regulated identifiers.

Administrators can only view the basic account metadata, like usernames and account creation timestamps, according to the MediaWiki administrative frontend. The admin interface doesn't let u see any sensitive personal data. The user table only contains the username, hashed password, timestamps, optional real name, and optional email, related to the database inspection on RockyDBServer. Even if a user tried to enter sensitive information, the system would have nowhere to store and go because all SPII-related fields are totally missing from the schema. UBNetDef SysSec comes to the conclusion that SPII cannot exist on the UBNetDef Wiki in its current configuration since the database structure does not support SPII fields and the software never requests, processes, or stores SPII.

SectionI (continued)

Evaluation of CIO Concerns, Threat Likelihood, and Risk Consequences

The CIO's claim that the UBNetDef Wiki poses a high degree of legislative or regulatory risk is considerably undermined by the fact that sensitive PII (SPII) is not present on the site. According to industry reporting, the majority of threat actors prioritize environments that store data with obvious monetary value, like payment cards, government identifiers, or vast amounts of customer records, because they are financially motivated (Verizon 2024). Compared to companies that handle SPII, UBNetDef SysSec determines that the probability of a motivated attacker purposefully targeting the UBNetDef Wiki is low because it only saves usernames, hashed passwords, and optional email or real name fields. The absence of sensitive PII (SPII) on the UBNetDef Wiki significantly undercuts the CIO's assertion that the site presents a high level of legislative or regulatory risk. Industry reporting indicates that most threat actors are financially motivated and prioritize environments that store data with clear monetary value, such as payment cards, government identifiers, or large volumes of customer records (Verizon 2024). Since the UBNetDef Wiki only saves usernames, hashed passwords, and optional email or real name fields, UBNetDef SysSec concludes that the likelihood of a motivated attacker deliberately targeting the site is low in comparison to organizations that handle SPII.

Some aspects of the database and supporting infrastructure could be vulnerable to information theft, manipulation, or denial of availability if basic security controls are misconfigured. For example, weak firewall rules, unpatched MediaWiki or database software, or exposed administrative interfaces could allow an attacker to alter Wiki content, steal hashed credentials, or temporarily disrupt service. In line with the trends highlighted by IBM X-Force—where credential harvesting and data theft represent some of the most common impacts—an adversary who compromises the UBNetDef Wiki would likely focus on abusing valid accounts, vandalizing content, or reusing stolen passwords against other sites rather than extracting SPII (IBM Security 2025).

Since the UBNetDef Wiki does not contain sensitive PII (SPII), the CIO's assertion that it presents a high level of legislative or regulatory risk is seriously called into question. Industry reports indicate that most threat actors are financially motivated and prioritize environments that store data with clear monetary value, such as payment cards, government identifiers, or large volumes of customer records (Verizon 2024). Since the UBNetDef Wiki only saves usernames, hashed passwords, and optional email or real name fields, UBNetDef SysSec concludes that the likelihood of a motivated attacker deliberately targeting the website is low when compared to organizations that handle SPII.

Section 2: Wazuh SIEM Review

Section II – Assessment of Wazuh as an Alternative SIEM Solution

UBNetDef SysSec finds that Security Information and Event Manager (SIEM) Wazuh is a suitable and cost-effective alternative to the SolarWinds SIEM proposed for the UBNetDef Wiki.

Wazuh is an open-source SIEM and extended detection and response (XDR) platform that offers endpoint security features for us to use, file integrity monitoring, real-time threat detection, and centralized log collection for use. Wazuh uses an open-source model and doesn't charge recurring licensing fees, in contrast to SolarWinds, a commercial enterprise SIEM that demands a hefty upfront bill for their license fee and continuous subscription costs. The platform is well-suited to the UBNetDef Wiki environment because it uses server-and-agent architecture that can get logs from web servers, Linux servers, and application services like MediaWiki and MariaDB.

Wazuh offers a number of essential features that satisfy the same high level security requirements as SolarWinds Security Event Manager. SolarWinds is meant to detect unwanted access, correlate security events, centralize log management, and notify administrators of malicious activity, which it does. But, these same objectives are accomplished by Wazuh's rules engine, which continuously examines gathered logs for questionable trends like recurring unsuccessful login attempts, attempts at privilege escalation, or unexpected process activity. Its intrusion detection capabilities find known attack methods, and its File Integrity Monitoring (FIM) and Security Configuration Assessment (SCA) modules identify unauthorized changes to configuration files and system binaries. Together, these features offer the UBNetDef Wiki the same useful risk-reduction results that SolarWinds is meant to offer, but without the significant financial burden.

By improving visibility into what is occurring on each host and throughout the network, Wazuh would potentially reduce the possibility of successful exploitation against the UBNetDef Wiki infrastructure. Because they don't have centralized monitoring and only examine logs after an incident has caused damage, which makes small environments frequently compromised. Events from UbuntuWebServer, RockyDBServer, and other hosts can be correlated almost instantly when Wazuh is installed on the UBNetDef Wiki network. This makes it possible to identify configuration errors, brute-force login attempts, and unusual access patterns early on, before an attack is successful. Because their actions are more likely to be noticed and looked into before they have a significant impact, the body of threat agents that are currently in existence and capable of attacking a basic Wiki platform becomes less effective.

Additionally, Wazuh fixes a number of existing or possible flaws in the UBNetDef Wiki environment, especially those pertaining to incident response and monitoring. Lack of centralized logging and alerting is still a vulnerability, even with properly configured firewall rules and system hardening: administrators might not detect suspicious activity until much later.

Wazuh bridges this visibility gap and improves the incident response process by combining logs, using detection rules, and producing alerts from different souces.. From a risk standpoint, permissive network rules and a weak or informal incident response workflow are equally vulnerable, and Wazuh directly mitigates this vulnerability. If a threat actor were to gain access to the database or web server, the detailed logs and alerts provided by Wazuh would shorten the time to detection and containment, limiting how far the attacker can progress.

Wazuh would facilitate quicker investigation and recovery in the case of a successful attack, thereby reducing the impact. An attacker could still try to reuse hashed passwords elsewhere, deface pages, or interfere with availability, even though Section I demonstrated that the UBNetDef Wiki does not store sensitive PII. The forensic information required to reassemble the attacker's activities, locate the compromised systems, and confirm that remediation measures were successful is provided by Wazuh's logs, correlation, and visualization features. This lessens downtime, limits the severity of any incident, and contributes to preserving wiki trust even in the event of a compromise.

In the event of a successful attack, Wazuh would try to lessen the impact by enabling speedier investigation and recovery. Although Section I showed that the UBNetDef Wiki does not store sensitive PII, an attacker could still attempt to deface pages, reuse hashed passwords elsewhere, or disrupt availability. Wazuh's log retention, correlation, and visualization features provide the forensic data needed to piece together the attacker's actions, identify the compromised systems, and verify that remediation efforts were successful. This helps to maintain wiki trust even in the case of a compromise, reduces downtime, and limits the severity of any incident.

Wazuh would mitigate the effects of a successful attack by facilitating quicker investigation and recovery. An attacker could try to deface pages, reuse hashed passwords elsewhere, or interfere with availability even though Section I demonstrated that the UBNetDef Wiki does not store sensitive PII. The forensic information required to piece together the attacker's actions, locate the compromised systems, and confirm that remediation efforts were successful is provided by Wazuh's log retention, correlation, and visualization features. This lessens downtime, limits the severity of any incident, and helps to preserve wiki trust even in the event of a compromise.

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Appendix

Figure 1 – MediaWiki Account Creation Page, This figure shows the user registration form, which only collects a username, password, and optional fields for email address and real name. No fields for Sensitive PII (SPII) exist in the registration workflow..... 3**Error! Bookmark not defined.**

Figure 2- Mediawiki Administrative User List, 2This figure displays the administrator's view of registered accounts. Only usernames and basic metadata (such as creation timestamps) are visible; no SPII can be accessed through the administrative frontend..... **Error! Bookmark not defined.**

Figure 3 MediaWiki Database User Table (RockyDBServer) , This figure shows the database schema and stored records inside the user table. Only minimal information is stored in the database: username, optional email, optional real name, hashed password, and timestamps. No SPII fields exist anywhere in the schema ..... 3.